



Republic of Rwanda
Ministry of Education



SWDML501

MACHINE LEARNING APPLICATION

APPLY MACHINE LEARNING FUNDAMENTALS

Competence

RQF Level: 5

Learning Hours



80

Credits: 8

Sector: ICT AND MULTIMEDIA

Trade: SOFTWARE DEVELOPMENT

Module Type: Specific

Curriculum: ICTSWD5001 TVET Certificate V in SOFTWARE DEVELOPMENT

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Purpose statement	This specific module describes the skills, knowledge and attitude required to Apply Machine Learning Fundamentals. This module is intended to prepare students pursuing TVET Level 5 in software development. Upon completion of this module, the learner will be able to Apply Data Pre-processing, Develop Machine Learning Model and Perform Model Deployment						
Learning assumed to be in place	Python programming, Mathematical analysis statistics and probability						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	30%	50%
	Practical work:			70%		70%	
	Group project and presentation	30%					
	Individual project /Work	40%					
				Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Apply Data Pre-processing	1.1 Environment is properly prepared based on system requirements.
	1.2 Data is properly manipulated based on the python libraries functionalities.
	1.3 Visualization results are properly interpreted based on its statistical analysis.
	1.4 Data cleaning is appropriately performed based on the provided dataset.
2. Develop Machine Learning Model	2.1 Machine Learning algorithm is properly selected based on the characteristics of the dataset.
	2.2 Machine Learning models are properly trained based on a training set of data.
	2.3 Machine Learning model performance is properly evaluated based on appropriate evaluation metrics
	2.4 Hyperparameters are properly finetuned based on evaluation results

3. Perform Model deployment	3.1. Deployment methods are clearly selected based on the requirements.
	3.2. Model file is properly integrated in the system based on the Deployment method (RESTful API guidelines.)
	3.3. Prediction responses are accurately delivered to the Clients based on the model insights.

Knowledge, Skills, and Attitude

Knowledge	Skills	Attitude
Description of machine Learning concepts	- Setting up a Machine learning environment - Extracting Dataset - Training Machine Learning model - Integration of Machine Learning model	- Quick learning - Creativity - Adaptability - Critical thinking

Learning outcomes	At the end of the module the learner will be able to: 1. Apply Data Pre-processing 2. Develop Machine Learning Model 3. Perform Model Deployment
Learning outcome 1: Apply Data Pre-processing	Learning hours: 30
Indicative content	
Description of Machine learning concepts ✓ Machine learning overview ✚ Definition ✚ Machine learning life cycle ✚ Machine Learning applications ✚ Advantages and disadvantages ✚ Difference between machine learning, artificial intelligence, and deep learning ✓ Types of Machine Learning	

- ✚ Supervised
- ✚ Unsupervised
- ✚ Semi-supervised
- ✚ Reinforcement
- ✓ Machine Learning tools
- **Preparing Machine Learning environment**
 - ✓ Installation of Python
 - ✓ Installation of Tools
 - ✓ Environment Testing
- **Data Collection and Acquisition**
 - ✓ Description of key terms
 - ✚ data
 - ✚ Information
 - ✚ dataset
 - ✚ Data warehouse
 - ✚ Big data
 - ✓ Identification of Source of data
 - ✚ IoT Sensors
 - ✚ Camera
 - ✚ Computer
 - ✚ Smartphone
 - ✚ Social data
 - ✚ Transactional data
 - ✓ Description of 6 V's of Big Data
 - ✚ Volume
 - ✚ Variety
 - ✚ Velocity
 - ✚ Veracity
 - ✚ Value
 - ✚ Variability
 - ✓ Description of Types of data
 - ✚ Structured data
 - ✚ Semi-structured data
 - ✚ Unstructured data
 - ✓ Gathering Machine Learning dataset
- **Interpret Data Visualization**
 - ✓ Description of data Visualization tools
 - ✚ Matplotlib
 - ✚ Seaborn
 - ✚ Plotly
 - ✚ Tableau
 - ✚ Power BI
 - ✓ Use Types of Data Visualization

- ✚ Scatter Plots
- ✚ Line Plots
- ✚ Bar Charts
- ✚ Histograms
- ✚ Box Plots
- ✚ Heat map
- ✓ Applying data Visualization Best Practices
- ✓ Interpreting visualizations results
 - ✚ Patterns and Trends
 - ✚ Context and Background
 - ✚ Correlations and Relationships
- **Perform Data cleaning**
 - ✓ Data cleaning overview
 - ✚ Definition
 - ✚ Purpose
 - ✚ Steps
 - ✓ Description of Characteristics of quality Data
 - ✚ Accuracy
 - ✚ Completeness
 - ✚ Consistency
 - ✚ Relevance
 - ✚ Validity
 - ✓ Data cleaning for inconsistencies rectification
 - ✚ Importance of data Cleaning
 - ✚ Data cleaning Techniques
 - ✓ Data normalization
 - ✚ Importance of data normalization
 - ✚ Data normalization Techniques
 - ✓ Data transformation
 - ✚ Importance of data transformation
 - ✚ Data transformation Techniques
 - ✚ Types of Data transformation

Resources required for the learning outcome

Equipment	● Computer, Projector, Storage Devices
Materials	● Internet
Tools	● Datasets, Python Distribution (CPython, Anaconda), Python IDEs (Jupyter Notebook, PyCharm, Visual Studio Code, or Spyder), Cloud jupyter notebook platform
Facilitation techniques	● Group discussion

	• Brainstorming • Practical exercises • Trainer guided
Formative assessment methods	• Written assessment • Oral questions • Performance assessment

Learning outcome 2: Develop Machine Learning Model	Learning hours: 30
Indicative content	
• Description of Machine learning algorithms and applications ✓ Supervised Learning Algorithms  Linear Regression  Logistic Regression  Decision Trees  Random Forest  Support Vector Machine (SVM)  k-Nearest Neighbors (KNN)  Naive Bayes ✓ Unsupervised Learning Algorithms  K-Means Clustering  Principal Component Analysis (PCA)  Hierarchical Clustering  Anomaly Detection ✓ Semi-supervised Learning Models ✓ Reinforcement Learning  Q-Learning	

- + Deep Q-Networks (DQN)

- ✓ Neural Network Architectures

- + Feedforward Neural Networks

- + Neural Networks (Artificial Neural Networks)

- + Convolutional Neural Networks (CNNs)

- + Recurrent Neural Networks (RNNs)

- **Selection of machine learning algorithm**

- ✓ Identify problem

- + regression

- + classification

- + clustering

- ✓ analyse type of data in a given dataset

- + Independent variable

- + Dependent variable

- ✓ Resource analysis

- + computational power

- + memory limitations

- ✓ choose machine learning algorithm to be used

- **Train machine learning model**

- ✓ Load a dataset

- + Using pandas

- + Using Numpy

- + Using Scikit-Learn

- + Using Seaborn

- + Using Requests and io

- ✓ Split dataset

- + train set

- + test set

- + validation set

- ✓ Initialize model

- ✓ Fit the training data into a model

- **Evaluation of machine learning model**

- ✓ prediction of result
 - ✚ on test data
 - ✚ on new data (unseen data)
- ✓ Visualize predictions
- ✓ Analyse evaluation metrics
 - ✚ Accuracy
 - ✚ precision
 - ✚ recall
 - ✚ F1 score
 - ✚ Mean Absolute Error (MAE)
 - ✚ Root Mean Squared Error (RMSE)
 - ✚ R Squared score
 - ✚ Adjusted R Squared score

- ✓ Model interpretation

- **Tuning Hyperparameters**

- ✓ Define Hyperparameters search space
- ✓ Choose performance metric
- ✓ Perform hyperparameter search
- ✓ Evaluate performance
 - ✚ Get the best parameters
 - ✚ Get the best model
 - ✚ predict on validation data
 - ✚ apply evaluation metrics on validation data
 - ✚ apply evaluation metrics on test data
- ✓ Resolve Potential Bias
 - ✚ Underfitting
 - ✚ Overfitting

Resources required for the indicative content

Equipment	● Computer, Projector, Storage Devices
Materials	● Internet
Tools	● Datasets, Python Distribution (CPython, Anaconda), Python IDEs (Jupyter Notebook, PyCharm, Visual Studio Code, or Spyder), Machine Learning

	Libraries (NumPy, Pandas, Scikit-Learn, TensorFlow or PyTorch, Matplotlib, Seaborn, Keras, SciPy), Cloud platform
Facilitation techniques	● Group discussion ● Brainstorming ● Practical exercises ● individualization
Formative assessment methods	● Written assessment ● Oral questions ● Performance assessment

Learning outcome 3: Perform Model Deployment	Learning hours: 20
Indicative content	
● Selection of model deployment method ✓ Description of model deployment methods + Definition + Benefits ✓ Identification of system specifications + Types of application (web application, mobile app, standalone program, Embedded system) + Technology (programming languages and frameworks) ✓ Identification of model specifications + Size of the dataset + Memory limitations + Computing power + Format of the model file(Scikit-Learn, TensorFlow SavedModel, ONNX, PyTorch PT) ● Integration of model file ✓ Integration goal + Predictions/Insights + Generate content + Data analysis	

- ✓ Compatibility
- ✓ Interpret API endpoint usage
 - ✚ Model serving method
 - ✚ Loading strategy
- ✓ Integrate with existing systems
- ✓ Identification data format: Establish the format for input data sent to the API
 - ✚ JSON
 - ✚ Form data
- ✓ Implement communication
 - ✚ Use HTTP requests
 - ✚ HTTP Responses to interact with existing system
- ✓ Testing thoroughly deployment
- ✓ Deploy to production
- ✓ Monitor performance
- ✓ Track API requests, response times, and model accuracy.
- **Delivering Prediction to the clients**
 - ✓ Integrating the API into application
 - ✓ Formatting the predictions
 - ✓ Handling errors

Resources required for the indicative content

Equipment	Computer, Projector, Storage Devices.
Materials	Internet
Tools	Datasets, Python Distribution (CPython, Anaconda), Python IDEs (Jupyter Notebook, PyCharm, Visual Studio Code, or Spyder), Machine Learning Libraries (NumPy, Pandas, Scikit-Learn,

	TensorFlow or PyTorch, Matplotlib, Seaborn, Keras, SciPy), Cloud platform, web application framework, web server
Facilitation techniques	• Group discussion • Trainer guided • brainstorming • Practical exercises
Formative assessment methods	• Written assessment • Oral question • Performance assessment

Integrated assessment

ABZ Company, a pioneering real estate software firm headquartered in Kigali, Rwanda, has been a market leader since its establishment in 2016. Their platform facilitates the seamless buying and selling of houses, catering to a substantial user base. However, recently, the company has been inundated with complaints from users regarding discrepancies in listed house prices. Users have expressed concerns that some sellers are setting prices that do not accurately reflect the value of the properties or the local market conditions.

Upon investigating these claims, ABZ Company discovered that certain sellers were listing properties either to gauge their current market value or due to inadvertent errors. Recognizing the importance of addressing this issue to maintain user trust and ensure fair transactions, ABZ Company has made the strategic decision to leverage machine learning technology.

In response to these challenges, ABZ Company is seeking to hire a skilled machine learning engineer to develop a robust pricing model and deploy it using APIs. This model will utilize various factors to accurately predict the price range of houses listed on the platform. By implementing such a solution, ABZ Company aims to enhance transparency, fairness, and accuracy in property pricing, thereby improving the overall user experience.

This work should be done in 8 hours

NB: Dataset and web application have to be provided by a trainer.

Resources

Equipment	Computer, Storage Devices
Materials	Python Distribution (CPython / Anaconda), Python IDEs (Jupyter Notebook / PyCharm / Visual Studio Code, or Spyder), Machine Learning Libraries (NumPy, Pandas, Scikit-Learn, TensorFlow or PyTorch, Matplotlib, Seaborn, Keras, SciPy), web application framework, web server
Tools	Internet, Dataset, web application

Assessable outcomes	(Assessment criteria based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Apply Data Pre-processing (30%)	1.1 The environment is properly configured according to specific functionalities.	Functionalities are specified			5
		The environment is configured.			5
	The data is properly acquired based on the problem requirements	Data are acquired from the given data source			10
	Data is appropriately pre-processed to obtain a dataset according to the intended model.	Data are pre-processed			20
Learning Outcome 2: Develop Machine	The features are properly engineered based on data analysis requirements.	Data features are engineered			10
		Model features are engineered			10

Learning Model (50%)	Machine learning algorithms are properly selected based on the dataset and problem requirements.	Machine learning models to be used are identified			5
	Model parameters are properly selected according to the algorithm.	Model parameters are selected.			5
	The model is correctly trained based on the algorithm and parameters.	Models are trained and tested.			20
Learning Outcome 3: Perform Model deployment (20%)	Metrics are appropriately selected to measure performance accuracy according to the trained model	Metrics are selected.			2
		Metrics are implemented			8
		Model parameters are tuned.			5
	The machine learning model is properly deployed into production using relevant deployment best practices.	The selected fit model is saved.			2
		The model is deployed to the web app.			3
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 50%					

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